Sharp Guarantees for Ensemble Voting

An elementary analysis of dependence — DSAA2011 Project 2

Project team — replace with actual members

The question and the contribution

For $K = 2m + 1$ fixed binary classifiers, let Z_i indicate correctness.

$$S = \sum_i Z_i, \qquad A = \mathbb{P}(S \geq m + 1).$$

- What do marginals and correlations determine about A?
- When must A exceed the common member accuracy p?
- Our focus: elementary sharp bounds, explicit attaining laws, and a finite-K threshold.

Sharp moment bounds and the insufficiency of pairwise information are classical. The project contributes a self-contained simplification and synthesis, not a claim of the first two-moment inequality.

Part I: opposite behavior with and without independence

$$A_{\mathrm{ind}} = \sum_{s=m+1}^{K} \binom{K}{s} p^s (1-p)^{K-s}, \qquad A_{-}^{\mathrm{marg}} = \max\left(0, \frac{Kp-m}{m+1}\right).$$

For $p > 1/2$, independent accuracy tends to 1, whereas the worst marginal-only accuracy tends to $2p - 1$.

The lower bound follows from $S \le m + (m+1)\mathbf{1}\{S > m\}$. It is attained using only counts m, K when $Kp > m$.

Unequal marginals: sort $p_{(1)} \ge \cdots \ge p_{(K)}$ and use

$$\max\left(0, \max_{r \ge m+1} \frac{\sum_{i=1}^{r} p_{(i)} - m}{r-m}\right).$$

Weights alone cannot improve the marginal-only minimax value beyond the best classifier.

Part II: pairwise independence is not joint independence

For $K = 3$, common $p, r = \mathbb{E}Z_iZ_j$, write $q = \mathbb{P}(111)$.

$$\max(0, 2r - p) \le q \le \min(r, 1 - 3p + 3r), \quad A = 3r - 2q.$$

At $p = 0.6$, $r = 0.36$ (every pair independent):

	Q_0	Q_1	Q_2	Q_3	A
Worst	0	.48	.24	.28	.52
Independent	.064	.288	.432	.216	.648
Best	.16	0	.72	.12	.84

Conditional on $S = s$, choose the correct subset uniformly. This verifies every marginal and pairwise moment in the table.

Replace 2^K configurations by one integer count

With equal p and equal r, randomly permuting voter indices preserves all constraints and the objective.

$$\mu = \mathbb{E}S = Kp, \qquad \nu = \mathbb{E}S(S-1) = K(K-1)r.$$

Conversely, uniform allocation within each count realizes every count law with these moments.

The feasible moment region is the polygon through $(s, s(s-1))$, $s = 0, \ldots, K$. If $b = \lfloor \mu \rfloor$,

$$2b\mu - b(b+1) \le \nu \le (K-1)\mu.$$

This is a finite symmetrization argument, not an assumption of an infinite iid mixture.

Two families of simple quadratic certificates

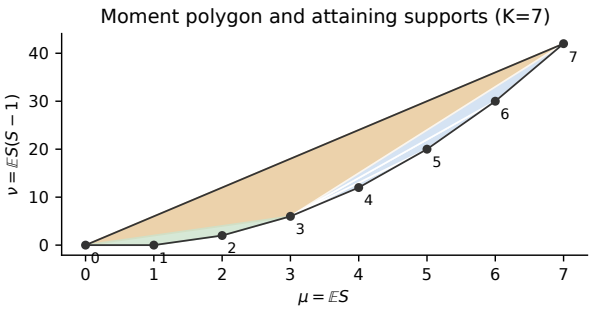
For every integer $s \in [0, K]$, all these functions are at most $\mathbf{1}\{s > m\}$:

$$0, \quad h(s) = \frac{s(s-m)}{K(K-m)}, \quad g_j(s) = 1 - \frac{(s-j)(s-j-1)}{(j-m)(j+1-m)}, \quad m+1 \le j < K.$$

- For $s \le m$, each function is nonpositive.
- For $s > m$, $h(s) \le 1$ and consecutive integers have nonnegative product, so $g_j(s) \le 1$.
- Taking expectations gives the exact lower bound B_K by maximizing these certificates.

The upper endpoint follows by complementing all votes: $S \mapsto K - S$.

Why the bound is attained: a picture is enough



- Green: losing counts only, so $A = 0$.
- Orange: support $\{0, m, K\}$ attains h.
- Blue: support $\{m, j, j+1\}$ attains g_j.

Every feasible point lies in a piece. Its barycentric coordinates are the required probabilities. No general LP duality is needed.

A one-line comparison gives constant-time evaluation

Set $d = \mu - m$, $v = \mathbb{E}(S - m)^2$, and $j = m + a$. Then

$$G_a = \mathbb{E}g_{m+a}(S) = \frac{(2a+1)d - v}{a(a+1)},$$

$$G_a - G_{a+1} = \frac{2[(a+1)d - v]}{a(a+1)(a+2)}.$$

If $d \leq 0$, every $G_a \leq 0$. Otherwise choose

$$a_* = \text{clip}(\lfloor v/d \rfloor, 1, m).$$

Only 0, $\mathbb{E}h$, and G_{a_*} need comparison. This is constant-time arithmetic given the two moments; it does not make reading a full pairwise matrix constant-time.

The exact threshold for beating a member

For $1/2 < p < 1$, define

$$x = \frac{Kp - (1-p)m}{p}, \quad j = \lfloor x \rfloor, \quad \theta = x - j,$$

$$\nu_* = (1-p)m(m-1) + p[2jx - j(j+1)], \quad \rho_* = \frac{\nu_*/[K(K-1)] - p^2}{p(1-p)}.$$

Necessary and sufficient, within the feasible range

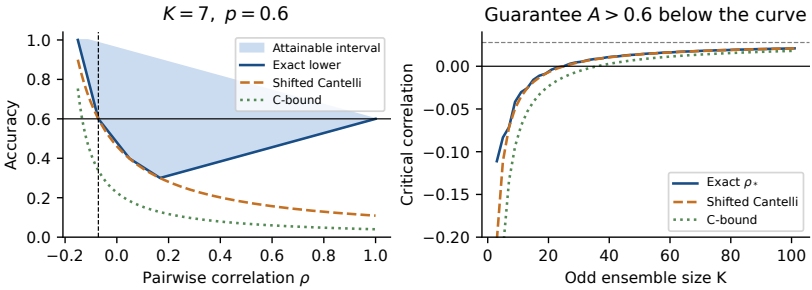
$$A_{\min} > p \quad \Longleftrightarrow \quad \rho < \rho_*.$$

At the boundary, put masses $1-p, p(1-\theta), p\theta$ on $m, j, j+1$. Accuracy is exactly p. Mixing with synchronized votes covers every correlation above the threshold without improving accuracy.

Separate the threshold correction from the integer correction

For $d = Kp - m$, the gain over the original C-bound threshold ρ_C is

$$\rho_* - \rho_C = \underbrace{\frac{p - 1/2 + 1/(4K)}{(K-1)p^2}}_{\text{correct failure threshold}} + \underbrace{\frac{\theta(1-\theta)}{K(K-1)(1-p)}}_{\text{integer support}}.$$



A concrete consequence: 27 pairwise-independent voters

For $p = 3/5$, $\rho = 0$, and $K = 2m + 1$,

$$\rho_* = \frac{m - 12 + (5/m)\mathbf{1}\{3 \nmid m\}}{18(2m+1)}.$$

	$K = 23$	$K = 25$	$K = 27$
Exact worst accuracy	.592	.600	.616

- The first strict guaranteed gain is at odd $K = 27$.
- Original C-bound first certifies it at 37.
- Shifted Cantelli also first certifies it at 27 in this example: the full improvement here is not a lattice effect.

Verification and what the result does not assume

- 702 count-LP cases; 41 full binary-LP cases; both endpoints checked.
- 720 threshold cases; integer boundaries, degenerate marginals, and attaining laws checked.
- Largest count-LP discrepancy below 3×10^{-14}.
- Proofs establish sharpness; experiments check implementations.
- Unequal pairwise constraints still use the general LP. Estimated moments require uncertainty regions.

Takeaway: pairwise information may not identify accuracy, yet it can certify an improvement. The integer count support makes that certification exact and elementary.

References and preparation for Q&A

Kuncheva & Whitaker (2003), *Measures of diversity*; Kuncheva et al. (2003), *Limits on majority vote accuracy*.

Germain et al. (2015), *Risk bounds for the majority vote*, Theorem 11 and Proposition 13.

Masegosa et al. (2020), *Second order PAC-Bayesian bounds*, Theorems 3 and 6.

Bertsimas & Popescu (2005), *Optimal inequalities in probability theory*; Boros & Prékopa (1989), *Closed form two-sided bounds*; Fontana & Semeraro (2021), arXiv:2101.07693.

Prepare to explain: why symmetrization is lossless, why the quadratics work only on integers, why the three-point law attains the threshold, and why population guarantees do not imply finite-sample confidence.

AI assistance: OpenAI Codex supported reading, derivation, code, and drafting. Team names and accurate individual contributions must be completed before submission.